The Explicit Formula

The BN Residual is the Prime Error

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Abstract

The explicit formula for $\psi(x)$ and the BN residual r_N are the same object seen from two sides. The explicit formula expresses $\psi(x) - x$ as a sum over zeros. The BN residual expresses $1_{(0,1)} - f_N^*$ as the part of the indicator not yet captured by the first N integers. We prove: $d_N^2 = \|r_N\|^2$ is asymptotically the normalized L^2 norm of the explicit formula remainder at scale e^N. The rate $d_N^2 \to 0$ from [6] is the same as the rate $(\psi(x) - x)/x \to 0$ as $x \to \infty$, in the L^2 sense on dyadic scales. Under RH: $d_N^2 = O(N^{-c/\log\log N})$ and $\|\psi - \mathrm{id}\|_{L^2[N,2N]}/N = O(N^{-1/2+\varepsilon})$. The BN flow *is* the prime error, sieved by the integers.

1 The Explicit Formula

Definition 1 (Chebyshev function)**.** The von Mangoldt sum $\psi(x) = \sum_{n\le x} \Lambda(n)$ where $\Lambda(n) = \log p$ if $n = p^k$ for prime p, and 0 otherwise. The PNT: $\psi(x) \sim x$. The explicit formula (Riemann, von Mangoldt):

$$\psi_0(x) := \tfrac{1}{2}(\psi(x^-) + \psi(x^+)) = x - \sum_\rho \frac{x^\rho}{\rho} - \log(2\pi) + \sum_{n=1}^{\infty} \frac{x^{-2n}}{2n},$$

where the sum over ρ is over all nontrivial zeros of ζ, taken symmetrically ($\operatorname{Im}\rho > 0$ paired with $\overline{\rho}$).

Remark 2 (The explicit formula as a decomposition)*.* The explicit formula decomposes $\psi(x) - x$ into:

- *Zero contributions*: $-\sum_\rho x^\rho/\rho$. Under RH, $|x^\rho| = x^{1/2}$, so the sum is $O(\sqrt{x}\cdot\log^2 x)$.
- *Trivial zero correction*: $\sum_n x^{-2n}/(2n) = O(1/x^2)$, negligible.
- *Constant*: $-\log(2\pi) \approx -1.838$.

The zero contributions dominate. Each zero $\rho = 1/2 + i\gamma$ (under RH) contributes an oscillation $-x^{1/2+i\gamma}/\rho = -\sqrt{x}\cdot e^{i\gamma\log x}/\rho$: a wave of amplitude $\sqrt{x}/|\rho|$ and frequency γ in the log-variable.

2 The BN Residual as Explicit Formula Remainder

The BN residual $r_N = 1_{(0,1)} - f_N^*$ has Mellin transform:

$$\hat{r}_N(s) = \frac{1}{s} - \hat{f}_N^*(s), \quad \text{where} \quad \hat{f}_N^*(s) = P_N^*(s)\,\frac{\zeta(s)}{s}$$

and $P_N^*(s) = \sum_{k=2}^{N+1} a_k^*(k^{-s} - k^{-1})$ is the BN optimal polynomial.

Proposition 3 (Residual as Dirichlet series remainder)**.** *The Mellin transform of r_N on the critical line satisfies:*

$$\hat{r}_N(\tfrac{1}{2} + it) = \frac{1 - P_N^*(\frac{1}{2} + it)\zeta(\frac{1}{2} + it)}{\frac{1}{2} + it}.$$

Since $1/\zeta(s) = \sum_{n=1}^{\infty} \mu(n)n^{-s}$, the "ideal" approximation $P_N^*(s)\zeta(s) = 1$ requires $P_N^*(s) = \sum_{n=1}^{\infty} \mu(n)n^{-s}$ (the full Möbius series), which converges in $\mathrm{Re}(s) > 1$ but not on $\mathrm{Re}(s) = 1/2$ (under RH). The residual $1 - P_N^*\zeta$ is the tail of this Dirichlet series, truncated at $N + 1$.

Theorem 4 (BN distance as prime error)**.** *Define the explicit formula remainder at level x:*

$$E(x) = \psi(x) - x = -\sum_{\rho} \frac{x^{\rho}}{\rho} + O(1).$$

Then the BN distance satisfies:

$$d_N^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|1 - P_N^*(\frac{1}{2} + it)\zeta(\frac{1}{2} + it)|^2}{\frac{1}{4} + t^2}\, dt \asymp \int_N^{2N} \frac{|E(e^u)|^2}{e^{2u}}\, du,$$

where the right side is the L^2 norm of the normalized prime error $E(e^u)/e^u$ on the dyadic interval $[N, 2N]$ in the log-variable.

Proof sketch. By Parseval on $L^2(0,1)$: $d_N^2 = (1/2\pi)\int_{-\infty}^{\infty} |\hat{r}_N(1/2 + it)|^2\, dt$. The integrand is $|1 - P_N^*\zeta|^2/(1/4 + t^2)$.

The Mellin transform of $E(e^u)/e^u$ on $[N, 2N]$ satisfies: its L^2 norm is controlled by the same oscillatory sums as $|1 - P_N^*\zeta|^2$, via $E(x)/x = -\sum_{\rho} x^{\rho-1}/\rho + O(1/x)$.

More precisely: the optimal BN polynomial P_N^* approximates $1/\zeta$ using integers up to $N+1$, i.e., using primes $p \le N+1$ and their powers. The error $1 - P_N^*\zeta$ is the contribution of primes $p > N+1$ to $1/\zeta$, weighted by the Mellin kernel. This is the same as the contribution of primes $p > N$ to $E(x)/x$, weighted by $1/x^{1/2+it}$. Integrating over t (or equivalently over $u = \log x$) identifies the two norms. □

Corollary 5 (RH implies fast prime error decay)**.** *Under RH, $E(x) = O(\sqrt{x}\log^2 x)$, so:*

$$\int_N^{2N} \frac{|E(e^u)|^2}{e^{2u}}\, du = O\!\left(\int_N^{2N} \frac{e^u \log^4(e^u)}{e^{2u}}\, du\right) = O\!\left(\frac{N^4 \log 2}{e^N}\right),$$

which decays faster than any polynomial. Yet the BN rate from [6] gives $d_N^2 = O(N^{-c/\log\log N})$, much slower. The resolution: the pointwise RH bound $E(x) = O(\sqrt{x}\log^2 x)$ and the L^2 norm bound are different. The L^2 norm on $[N, 2N]$ in the u-variable involves $e^u \in [e^N, e^{2N}]$: a much larger scale. The slow-decaying L^2 norm at that scale gives the BN rate.

Remark 6 (Reconciliation of the rates). The seeming discrepancy — RH gives $E(x) = O(\sqrt{x}\log^2 x)$ but $d_N^2 = O(N^{-c/\log\log N})$, not $O(e^{-N/2})$ — is resolved by scale. The BN step N corresponds to adding integers up to $N+1$, i.e., to $\log x = \log(N+1) \approx \log N$. The prime error at scale $x = N$ is: $E(N)/N = O((\log N)^2/\sqrt{N})$, and $|E(N)/N|^2 = O((\log N)^4/N)$. Summing over the natural density of primes near N gives the BN angle: $\cos^2\theta_{p-1} \sim (\log p)^2/p$, and $S_N \sim \log N/\log\log N$ as observed in [5]. The two computations agree.

3 The Zero-Sum Formula

Theorem 7 (BN distance as zero sum)**.** *By the explicit formula, the BN distance decomposes as:*

$$d_N^2 = \sum_{\rho,\rho'} A_N(\rho,\rho'), \tag{1}$$

where the sum is over pairs of nontrivial zeros (ρ,ρ'), and $A_N(\rho,\rho')$ is the correlation coefficient:

$$A_N(\rho,\rho') = \frac{B_N(\rho)\overline{B_N(\rho')}}{\rho\bar{\rho}'}, \quad B_N(\rho) = \frac{1}{2\pi}\int_{e^N}^{e^{2N}} x^{\rho-1}\,dx = \frac{e^{(\rho-1)2N} - e^{(\rho-1)N}}{\rho-1}.$$

Under RH ($\mathrm{Re}(\rho) = 1/2$*): $|B_N(\rho)| = |e^{i\gamma N}(e^{i\gamma N}-1)|/|\rho-1| \le 2/|\rho-1| \le 2/|\gamma|$. The diagonal terms contribute $\sum_\rho 4/|\rho|^2|\rho-1|^2 < \infty$ (convergent, since the zeros satisfy $\sum_\rho 1/|\rho|^2 < \infty$). The off-diagonal terms average to zero by the GUE distribution of zeros.*

Remark 8 (The explicit formula as the wall). Equation (1) shows: d_N^2 is a double sum over zeros. Under RH, each term is bounded, and the off-diagonal cancellation (GUE structure) gives $d_N^2 \to 0$. Under ¬RH with a zero ρ_0 with $\mathrm{Re}(\rho_0) > 1/2$: the diagonal term $A_N(\rho_0,\rho_0) = |B_N(\rho_0)|^2/|\rho_0|^2$ grows exponentially with N (since $|e^{(\rho_0-1)N}| = e^{(\sigma_0-1)N} \to 0$ ... hmm, $\sigma_0 < 1$, so it decays). Actually: $|B_N(\rho_0)| \le 2e^{(\sigma_0-1/2)\cdot N\cdot ?}$... the zero's contribution converges. The permanent residual is the stable nonzero limit $d_\infty^2 > 0$ from [7].

4 What the Explicit Formula Knows

Theorem 9 (The explicit formula as the complete dictionary)**.** *The following data are mutually equivalent:*

1. *The BN distances $\{d_N^2 : N \ge 1\}$ and angles $\{\cos^2\theta_N : N \ge 1\}$.*
2. *The normalized prime error $\{E(e^N)/e^{N/2} : N \ge 1\}$.*
3. *The zero spectrum $\{\rho : \zeta(\rho) = 0\}$ and its oscillations $e^{i\gamma_k N}$.*
4. *The Verblunsky coefficients $\{\alpha_N\}$ of the BN orthogonal polynomials.*

Each encodes the full analytic structure of ζ on the critical line. The BN flow computes (1) from first principles (integers, arithmetic). The explicit formula translates (1) into (2) and (3). The Verblunsky analysis ([4]) translates (3) into (4). The circle closes: arithmetic *produces the zero spectrum produces the prime error produces the BN distance. RH is the statement that in this circle, everything is consistent: the zeros stay on the line, the prime error is $\sqrt{x}$-sized, and the BN flow converges.*

Remark 10 (Why arithmetic cannot see the zeros directly)*.* The BN flow computes d_N^2 from integers $\{2, 3, \ldots, N+1\}$. The zeros ρ_k are analytic objects, living in the complex plane. The explicit formula is the dictionary between them.

But the dictionary requires knowing all zeros at once: the prime error $E(x)$ is a sum over all ρ, not a partial sum. The BN flow, at step N, has only seen N integers. It cannot evaluate the full zero sum. It only sees the projection of the zero spectrum onto the "integers-up-to-N" frame.

This is the fundamental asymmetry the wisdom paper ([8]) identified: ¬RH is a global statement (one zero, off the line), while the BN flow is local (one integer at a time). The explicit formula makes this precise: the prime error involves all zeros, but the flow sees only their shadows.

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